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Research Paper

Improving organizational processes in healthcare through simulation-driven resource allocation: Methodology and real-world case study

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ABSTRACT

The global rise in the aging population presents significant challenges to healthcare systems worldwide, which thus need more efficiency and effectiveness. Healthcare systems deliver services through processes that encode (inter)national regulations and protocols for patient treatment. It follows that efficiency and effectiveness require improvement of healthcare processes. This paper introduces a novel methodology leveraging process mining and simulation to improve healthcare organizational processes through data-driven resource allocation. Traditional qualitative analyses and queuing theory offer limited scope for complex, multi-activity processes.

In contrast, process simulation enables process improvement but requires a realistic simulation model; otherwise, the analysis may optimize an assumed process rather than the real one. This paper reports on a data-driven simulation methodology that builds on process mining techniques: process mining puts aside the subjectivity of process actors and focuses on the objectivity of transactional data recorded by information systems. This enables us to discover process simulation models that mimic real behavior. Simulation models support process improvement through the evaluation of alternative *what-if* scenarios, which can be tested without disrupting live operations. The methodology has been applied to the emergency department of an Italian hospital, focusing on reducing patient waiting times. Simulations involving a careful addition of medical staff demonstrated a potential substantial reduction in waiting times (88%) with a modest cost increase (7%–8%). Furthermore, the improved process exhibited enhanced resilience to surges in patient arrivals, highlighting how improved processes guarantee higher preparedness in front of emergencies.

1. Introduction

Recent reports, e.g. from World Health Organization (WHO) [1] and from the World Bank [2], are clearly indicating that every nation is witnessing a rise in both the number and proportion of older adults within their populations. By 2030, one in six people globally will be aged 60 or older. The population in this age group is projected to grow from 1 billion in 2020 to 1.4 billion. Looking ahead to 2050, the number of people aged 60 and over is expected to double, reaching 2.1 billion. Additionally, the population of those aged 80 and above is anticipated to triple, increasing to 426 million by mid-century. This trend has originally been seen in western world but has recently been observed world-wide.

"All countries face major challenges to ensure that their health and social systems are ready to make the most of this demographic shift" [1]: the healthcare costs are destined to steadily grow, and a smaller share of population can be employed to provide healthcare support.

Healthcare administrations deliver services through processes that codify rules and regulations, and actions need to be taken on these processes to improve their efficiency and reduce the costs to tackle with the aging-population challenge. The improvement of healthcare processes would also reduce the errors, and thus enhance the patient outcomes, with better diagnosis, treatment, and overall care, reducing complications and mortality rates. This improvement would also positively affect the workforce well-being, with reduction of stress and burnout among healthcare professionals. Last but very importantly, improving healthcare processes would increase preparedness and resilience for emergencies, whose importance has been pointed out by the COVID-19 pandemic: the management of crises (pandemics, natural disasters, mass casualties, etc.) can be better dealt with through improved processes.

These processes, particularly in clinical settings, are often complex and can vary accordingly to several factors, situations, and patient characteristics. Therefore, their improvement is not straightforward.

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This paper's goal is to provide a methodology in which processes are improved, and the extent of this improvement is aimed to be quantified. Therefore, we exclude the use of qualitative-analysis techniques, such as Value-Added & Waste Analysis, Root-Cause Analysis, or Pareto Analysis [3]. Instead, we aim at quantitative analyses, which can be based on the applications of queuing theory or business process simulation [4]. Queuing-based analyses are excellent for capacity problems, which are common and a key driver of process redesign. However, a number of limitations exist, as pointed out by Dumas et al. [3]. They can be used to analyze waiting times (and hence processing times) but cannot focus on cost or quality measures. It is certainly suitable to analyze a single activity at a time that can be performed by resources in one single pool, but fails when aiming at analyzing end-to-end processes consisting of multiple activities performed by multiple resource pools. **Process simulation** provides a means of overcoming these limitations [3–5].

Process simulation refers to techniques that mimic the dynamic behavior of a process using different aspects. These techniques can be used to conduct *what-if* scenarios, avoiding the need for building or disrupting the real-world system. Through the analysis of different *what-if* scenarios, it is possible to assess the impact of the changes onto the processes, protocols, and norms leading to the initial/current process.

Process simulation requires a model that encodes the process being analyzed and needs to be coupled with parameters that specify the run-time simulation aspects (arrival rate, activity-duration distributions, etc.). Traditionally, these deliverables have been provided through interviews and questions with process actors and stakeholders, based on their subjectivity, ignoring objective insights that actual process transactional data can provide. Subjectivity in developing and employing simulation models can lead to theoretical process improvements which do not correspond to reality.

Process Mining is a discipline that bridges Artificial Intelligence and Data Mining techniques with process modeling, aiming primarily at uncovering how processes are actually executed [6]. Instead of relying on interviews, workshops, or theoretical models, which can be subjective or outdated, process mining analyzes the digital footprints left behind in IT systems to understand how processes actually run and to derive accordant models and run-time characterization of their executions. These digital footprints are represented as event logs, each recording the executions of instances of a certain process.

This paper brings forward a methodology that leverages process mining to discover the model of the process to simulate, along with the run-time characterization of the different process perspectives on activities, resources, and time. The methodology takes into account of several healthcare process characteristics, facing relevant, open challenges (cf. discussion in Section 3).

The methodology was concretely applied to the emergency department of an Italian hospital, with the aim of improving patient waiting times both before and during treatment. The methodology includes a validation phase to assess a sufficient level of realism of the simulation model, which was positively confirmed for the case study in question. The improvement attempt was carried out by simulating various *what-if* scenarios based on the addition of resources in different roles—such as specialists, nurses, and laboratory technicians. Since hiring resources for certain roles is more expensive than for others, the potential benefits must be weighed against the associated costs. These benefits are primarily reflected in the reduction of waiting times for activities requiring specific roles. Ultimately, this means that a trade-off exists that must be carefully balanced.

The results of the application of the methodology on the case study show that the waiting time can be potentially improved by approximately 88% compared to the original time, with a limited cost's increase of around 7%–8%. A simulation-driven analysis of the waiting times in case of various increases of arrival rate confirms that healthcare process improvement is also beneficial for the preparedness in case of emergencies: the patient's waiting time increases less than half when the patient arrival rate doubles, if compared with the original process.

The methodology presented in this paper is particularly well suited for **organizational healthcare processes**, i.e., processes that are structured, standardized, and primarily concerned with resource management and routing of patients through organizational units (as opposed to highly flexible, knowledge-intensive clinical workflows) [7]. For such processes our data-driven simulation discovery approach effectively captures control-flow, temporal, and resource perspectives from event logs, supporting realistic *what-if* analyses and resource optimization.

The structure of the paper is as follows. Section 2 introduces key concepts in the fields of Process Mining and Process Simulation. Section 3 then discusses open process mining challenges in the healthcare domain, positioning our research with respect to the current state of the art. Section 4 presents our methodology in a step-by-step manner. Section 5 describes the successful application of the methodology to the emergency department of an Italian hospital. Section 6 discusses potential threats to validity, both in relation to the proposed methodology and to the conducted case study, and identifies possible directions for future research. Section 7 reviews relevant literature on the application of Process Mining and Process Simulation in the healthcare domain. Finally, Section 8 summarizes the main findings and concludes the paper.

2. Background

2.1. Process mining

Process Mining is an emerging scientific discipline that aims to improve operational processes through the systematic use of process' event data [6]. The novelty of Process Mining compared with the traditional process intelligence lays in leveraging on the objectivity of the event data in place of human subjectivity (e.g. retrieved via questionnaires, interviews, etc.). This explains why Process Mining is considered as the missing link between Process and Data science: the aim remains to improve the operational processes, but focusing on the *real* processes, as described in the data, instead of the *supposed* processes, which would come from the subjective views of the process' actors.

2.1.1. Event logs

Process' event data are represented as **event logs**. Table 1 shows a tabular representation of an event log: each row represents an *event*, which captures a specific execution of an activity within a *case* (i.e., an instance of the process). Column headers represent attributes associated with events, i.e. a table field specifies the given column's attribute value of the row's event. Mandatory attributes are *Case Identifier*, which allows identifying to which case the event refers, the *Timestamp* when the event occurred, and the *Activity* that was executed. Other typical attributes are the *Resource* that executed the activity and *Transactional information* indicating the activity's state. Using transactional information, one can determine the duration of the activities. As an example, the execution of activity *Lab Service* for case 002083 started at 1:30 on 15 March, and completed at 2:20, and hence, took 50 min.

2.1.2. Process models

Aside from event logs, another typical artifact of process mining techniques is a **process model**. A process model is a *description* of how a process ought to be executed. Compared with a textual description in natural language, a process model encodes (or should encode) the set of allowed executions in a more formal way. Using a formal process modeling notation has several advantages, i.e., it allows us to assess various quality properties of the model, e.g., absence of deadlocks. It

Table 1
Event log example based on the emergency department case study in Section 5. Each row represents an event with its attributes. The sequence of events sharing the same *Case Id* is a trace, i.e. the sequence of events of a single patient during the process.

Case Id	Activity	Timestamp	Lifecycle	Age	Access type	Urgency code	Pathology
...	...	...	...	...	...	...	...
002082	Access	Mar 15, 00:17	Complete	49	Autonomous	Blue	Skin rash
002082	Triage	Mar 15, 00:17	Complete	49	Autonomous	Blue	Skin rash
002082	Visit	Mar 15, 01:26	Complete	49	Autonomous	Blue	Skin rash
002082	Discharge	Mar 15, 01:27	Complete	49	Autonomous	Blue	Skin rash
002082	Exit	Mar 15, 01:28	Complete	49	Autonomous	Blue	Skin rash
002083	Access	Mar 15, 00:48	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Triage	Mar 15, 00:48	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Visit	Mar 15, 01:26	complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Blood sampling	Mar 15, 01:30	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Lab service	Mar 15, 01:30	Start	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	CT request	Mar 15, 01:30	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	CT accept	Mar 15, 01:32	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	CT execution	Mar 15, 01:55	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	CT reporting	Mar 15, 02:06	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Lab service	Mar 15, 02:20	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Medium short stay unit	Mar 15, 02:22	Start	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Medium short stay unit	Mar 15, 11:10	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Discharge	Mar 15, 11:10	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
002083	Exit	Mar 15, 11:48	Complete	72	Ambulance (no doctor)	Yellow	Mild TBI
...	...	...	...	...	...	...	...

Fig. 1. Example process model in BPMN notation, which represents the same activities as presented in the event log in Table 1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

also allows software tools to automatically reason about the modeled process behavior.

Several modeling notations have been put forward [6]. However, the current de-facto standard in industry is the Business Process Model and Notation (BPMN) formalism, which forms a trade-off between being mathematically grounded and human interpretable notation [3]. Fig. 1 illustrates a simple model, describing the same process behavior as captured in Table 1. This is not the real process model used in the case study discussed in Section 5, but is only a toy example to showcase the main constructs of BPMN.

The model describes a few fundamental constructs, often used when modeling a process. The green circle represents the starting point of the process, whereas the orange circle represents the end point. Each rounded rectangle represents an activity that we are able to execute within the process, e.g., the *visit* activity. The arcs represent the general behavioral flow of the model, e.g., the first activity is *access*.

The diamond-shaped operators represent behavioral control-flow constructs. The diamonds containing a ×-symbol represent an *exclusive choice*, i.e., either one of the outgoing/incoming arcs is followed. The diamonds containing a + -symbol represent *concurrency*: all outgoing/incoming arcs (and branches) are followed and executed completely independently/concurrently. The + -symbols with multiple incoming arcs are synchronization points. We refer the reader to [3] for a more elaborate introduction.

2.1.3. Process discovery

While Process Mining can be used to assess the compliance of process executions w.r.t. a set of norms and protocols or to provide run-time operational support, the lion's share of attention is certainly about process discovery [6].

Process discovery aims at derive a process model from an event log, e.g. in the form of a BPMN model. Several process-discovery algorithms exist in literature [6], where each algorithm has its representational bias: some can only discover models with specific constructs and may not capture all observed behavior. Algorithms also differ in their ability to separate legitimate behavior from rare or noisy events, incorporating primarily the former. This distinction is especially relevant in healthcare, where most behavior follows established medical protocols and only a small portion consists of exceptions. Focusing on the mainstream behavior is important when the goal is to improve healthcare process performance, such as reducing costs, minimizing waiting times, and increasing satisfaction for both professionals and patients.

Process Mining also provides metrics to assess the quality of a model. The main metrics used in the literature are *fitness*, *precision*, and *generalization* [6], whose computation requires a reference event log to be computed. Fitness measures the average level of compliance of the traces of the reference event log with the process model. Precision measures to which extend the process model do not allow process executions that are not observed in the reference event log. However, while we desire the model to be precise, we also aim at a model that is capable to fairly generalize beyond the observed behavior: an event log typically records only a portion of the full range of behavior that a process can exhibit, while a model aims to represent the entire behavior set. The latter is not measured by precision: metrics *generalization* aims to quantify how much the model is able to generalize beyond the behavior set that has been observed. For further information and formalizations of these metrics, the reader is referred to [6].

The discovery algorithms mentioned above focus on discovering a process model that defines the sequences of activities that are allowed by the process. However, a process incorporates other perspectives, mainly on resources, decisions, and time. These perspectives can also be discovered by analyzing event logs. An extensive discussion on the discovery of these other perspectives is provided in [8]. The remainder provides a short overview of those employed to construct business simulation models.

Burattin et al. [9] focus on the *resource perspective*: their technique is capable to determine the roles within the organization and to associate resources to these roles. Similarly, Camargo et al. [10] addresses the discovery of the resource perspective by clustering resources with similar activity execution profiles to identify resource pools and assign activities accordingly.

The decision perspective focuses on determining the conditions on the process-instance attributes (e.g., the patient's triage code, age, or suspected pathology) that drive the choices of the process paths (e.g., whether or not to perform a blood test, a certain type of visit, or imaging analysis). Correctly modeling these decisions is crucial for an understanding of the process, and even more for its accurate simulation [8,11].

The time perspective focuses on discovering the distribution of the activity durations, but also on evaluating potential bottlenecks [8]. Activity durations are rather easy to compute by computing the difference between the event indicating the completion of activities and the event indicating the starting. See, e.g., the two events related to the *lab service* activity in Table 1. From the individual activity durations, the best distributions can be easily determined using standard statistical techniques. Similarly, bottlenecks can be computed by considering the timestamp difference between the completion of an activity and the starting of the next.

Recently, object-centric process mining has been proposed to represent events associated with multiple object types and their relationships [12], providing a more natural view of complex interacting entities such as patients, lab tests, and imaging orders. It also shows promise for process simulation, as it supports the modeling of multi-entity scenarios [13,14]. However, current object-centric process simulation techniques are still in their early stages and face methodological and tooling limitations; therefore, this work focuses on classical process mining approaches based on the single-case notion.

2.2. Process simulation

Process simulation is one of the most widely used and supported techniques for quantitatively analyzing process models [3]. The core concept behind process simulation involves the use of a simulator to generate numerous hypothetical instances of a process. The simulator's output typically includes simulation logs, similar to event logs, along with key statistics, such as the averages of process-execution times, waiting times, resource utilization, or average costs.

Process simulation requires analysts to provide a simulation model $\mathcal{M} = (\mathcal{N}, \mathcal{P})$, consisting of a control-flow process model $\mathcal{N}$, such as in the BPMN modeling language, and a set of simulation parameters $\mathcal{P}$. Parameters $\mathcal{P}$ define the run-time characterization of the process simulation:

1. the probability distribution of activity durations,
2. the probabilities to follow each of the alternative branches at the exclusive choice,
3. the probabilities distribution of the arrival rate of new process instances,
4. the working hours,
5. the costs,
6. the roles of the resources involved in the process,
7. the assignment of resources to different roles where a resource can play different roles,
8. the assignment of the activity authorizations to roles (namely, the roles that a resource needs to play to be allowed/able to perform a certain activity).

Process simulation is certainly useful to simulate the current process, but is even more valuable for *what-if* analysis, which refers to simulate alternative scenarios, also referred to as *what-if* scenarios. These alternative scenarios are obtained by altering the process model and/or varying the simulation parameters. A *what-if* analysis consists of formulating alternative scenarios, simulating them, and measuring the impact on a certain Key Performance Indicator (KPI) of interest (e.g., the patient's waiting times). By inspecting these alternative scenarios, process analysts formulate realistic hypothesis of process improvement. The advantage of this analysis type over conducting on-the-field experiments resides in a significant reduction of the risks. A *what-if* scenario that yields very poor KPI values may, e.g., put in danger the hospital's patients, leading to deaths, or irreparable physical or psychic damages.

The problem can be formally formulated as a stochastic optimization problem aiming at finding the best simulation model $(\mathcal{N}, \mathcal{P})$ that minimizes the expected value of the objective function $f(\mathcal{N}, \mathcal{P})$ that encodes the KPI of interest. This might suggest that the formulation of an opportune Mixed-Integer Linear Program (MILP) could be an alternative. However, this is not really feasible: MILPs do not account for stochasticity, neither is it practical feasible to encode the constraints imposed by a process model $\mathcal{N}$ as a set of MILP constraints, especially when the model allows for executions that are arbitrarily long.

Compared with other techniques for quantitative analysis, first and foremost queuing analysis, process simulation is capable to analyze the single activities of an end-to-end processes and improve on them singularly, also considering the different roles and resources. Conversely, queuing analysis is only suitable to analyze the process as a whole, assuming it to be all performed by single resources [3]. Furthermore, queuing analysis can be used to analyze and improve on waiting times, but cannot be used to improve on cost or quality measures [3].

The simulation results are only as trustworthy as the underlying simulation model. An inaccurate model would produce unrealistic outcomes, leading to conclusions that may appear valid in theory but not in practice. This article aims to reduce subjective assumptions by deriving the simulation model and parameters directly from event data recorded in process logs. This ensures that the analysis and optimization are grounded in the observed behavior of the real process. While the evaluation is performed in a simulated setting, the approach provides an objective evidence-based foundation for assessing potential improvements prior to real-world implementation.

Section 2.1.3 indicates that many techniques exist in literature that focus on the discovery of the process' perspective related to control-flow, resource, and time. These discovery techniques are leveraged on in this paper to build simulation models and to use them for process improvement.

3. Our contribution to the open challenges

Process mining applications in the healthcare domain face several challenges to be dealt with [15]. Building on these insights, our work contributes to tackling several of the open challenges identified by Munoz-Gama et al. [15], through a methodology tailored to the specific needs and constraints of healthcare environments. Our proposed methodology aligns with several of these challenges:

C1: Design Dedicated/Tailored Methodologies and Frameworks We developed a methodology for healthcare scenarios, which integrates simulation and process mining to automate improvements. Its design reflects key features of organizational healthcare processes, such as urgency-driven workflows, variable patient arrivals, and time-critical resource allocation, identified in collaboration with healthcare stakeholders. Note that the underlying principles of the methodology, namely simulation-driven resource allocation and process improvement, might be valid in other domains with similar organizational and resource constraints, although we do not have a supporting case study that can concretely showcase the methodology's application to other domains.

C2: Discover Beyond Discovery We used simulation to go beyond traditional discovery by incorporating enhancement and conformance checking.

C4: Deal with Reality We assessed the realism of generated models using state-of-the-art accuracy metrics [26], ensuring fidelity to observed behavior and compliance with medical protocols.

Table 2
Overview of process simulation approaches in the Healthcare domain, and their attributes aligned with key challenges identified in [15]. "+" indicates that the corresponding work fully addresses the challenge, "+/−" indicates partial consideration. Empty cells indicate that the challenge is not addressed. Last row highlights our method.

Reference	Data quality improvement (C6)	Accounting for patient characteristics (C8)	Model accuracy assessment (C4)	Process optimization (Part of C1)
Guastalla et al. [16]				+
van Hulzen et al. [17]	+	+	+/−	
Halawa et al. [18]		+/−		+
Tamburis et al. [19]			+/−	
Antunes et al. [20]				+
Johnson et al. [21]	+	+	+/−	
Kovalchuck et al. [22]	+/−	+/−	+/−	
Abohamad et al. [23]			+/−	
Mans et al. [24]			+/−	
Ahmed et al. [25]				+
Our methodology	+	+	+	+

C5: Do It Yourself (DIY) Healthcare professionals were engaged throughout the methodology design to ensure practical relevance and usability, fostering the adoption of process mining tools in the field.

C6: Pay Attention to Data Quality Recognizing the critical impact of data quality, especially timestamp granularity and missing values, we implemented a preprocessing phase. This includes a novel method for estimating missing start timestamps, described in Appendix.

C8: Look at the Process through the Patient's Eyes Our simulations incorporate patient characteristics to reflect real-world variability. Optimization efforts emphasize patient-centric improvements, such as reducing waiting times, thus enhancing care pathways.

Table 2 compares our approach with existing simulation-based methodologies in healthcare (cf. Section 7), focusing on how each of these approaches addresses selected challenges.

The comparison is conducted at the conceptual and methodological level, as most existing works differ substantially in purpose and scope, often focusing on process modeling, monitoring, or analysis rather than optimization [17,19,21–24]. Conversely, methods such as those by Guastalla et al. [16], Halawa et al. [18], and Antunes et al. [20] primarily focus on model construction or simulation design, without incorporating data quality improvement, model accuracy validation, or patient-specific variability. Similarly, Ahmed et al. [25] rely on models that are manually built and not extracted from the data: the introduction of this stakeholders' subjectivity can lead to situations in which improvement is made on the supposed processes, which are detached from reality. Given these fundamental differences in objectives and methodological scope, a direct experimental comparison would not yield meaningful insights.

4. Healthcare process improvement via simulations

This section presents our data driven improvement methodology for healthcare processes. The methodology is designed mainly for organizational healthcare processes, where resource allocation decisions and predictable control-flows are primary levers for improvement.

The methodology is depicted in Fig. 2, where the phases are outlined: *Process Modeling (A)*, *Model Accuracy Assessment (B)* and *Process Optimization (C)*. The input is an initial event log $\mathcal{L}$, that, after an *Event Log Preparation* step, is used for discovering a simulation model, with user support if needed. In the second phase, the accuracy of the simulation model is assessed via measures that compare simulated event logs with the actual one through several perspectives. If the simulation model is not accurate, the previous modeling step is repeated by repairing the discriminated characteristics; otherwise, it is ready to be used in the optimization phase. Finally, we use the simulation model to obtain simulated event logs, from which statistics are derived to guide optimization in changing the simulation parameters. This phase is repeated until a stop criterion is reached or until no further improvement is possible, returning a simulation model that reflects the optimal process configuration.

The approach leverages Process Mining and Process Simulation approaches for building an accurate process simulation model (cf. Section 4.2). The model accuracy assessment is based on the measure proposed by Chapela-Campa et al. [26] (cf. Section 4.3). Once the simulator is built and the accuracy is assessed, it can be used by an optimization algorithm to solve a determined optimization problem (cf. Section 4.4).

4.1. Event log preparation

The starting point is the creation of a proper event log to use as input to discover a simulation model. As discussed in Section 2.1.3, the discovery of the distribution of the activity duration requires logs to contain the events that indicate the starting and completion of each activity execution.

Healthcare information systems do not always record the starting and completion timestamps of activities, especially when referring to visits and tests. In these case, typically the timestamps are the completion and taken from the medical reports. Even when present, these timestamps often lack sufficient granularity, e.g. only recording the date of the activities but not the exact time [27–30]. When some timestamps are missing or too course, we need to preprocess the event log, and to estimate to refine them.

Fracca et al. [31] introduced a new framework to estimate timestamps when they are not present or are not reliable. We have improved the framework, which is now capable to scale with larger event logs and can estimate the timestamps more precisely. The improved framework is discussed in Appendix. Note that the framework is based on simulation and thus requires a simulation model, of course without the characterization of the distributions of the activity duration, which the framework aims to facilitate.

4.2. Process simulation discovery

Once the event log has been prepared, our methodology aims to discover a process simulation model $\mathcal{M} = (\mathcal{N}, \mathcal{P})$, from the event log, replacing the human subjectivity in creating model with the objective behavior observed in the event log (cf. discussion in Section 2.2).

In our methodology, the process model $\mathcal{N}$ is mined from the event log using the process discovery techniques discussed in Section 2.1.3.

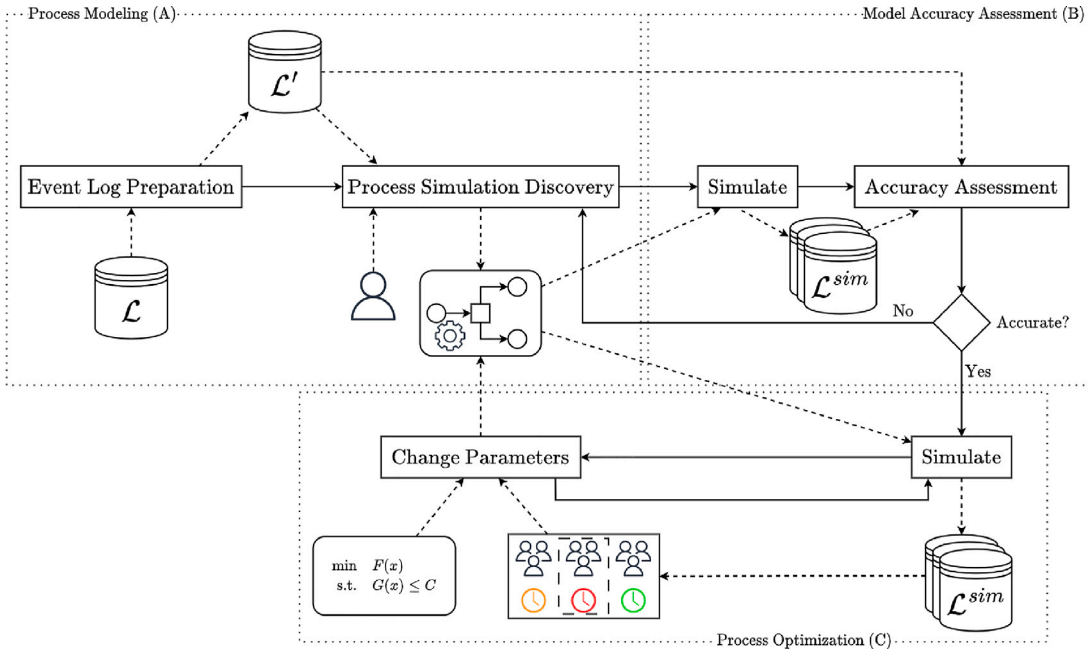

Fig. 2. Healthcare process improvement via simulations schema. In an initial step the event log is firstly prepared to use it for process discovery techniques, and optionally including user validation, for building a process model then enhancing it with multi perspective parameters. Step B assesses the accuracy of the simulations. Finally, the last step uses simulations for providing optimal configurations. Rectangles and solid arrows represent the method workflow, while elements connected with dashed lines indicate input and output objects.

The simulation parameters $\mathcal{P}$ are derived from the event log whenever relevant information is available, for example, probabilities of activity executions, arrival rates, and temporal distributions (cf. Section 2.2). When such information is not present in the event log, it can be provided directly by domain stakeholders. This was the case in the emergency department study presented in Section 5, where resource-related data were not available in the event log. Therefore, healthcare stakeholders provided role definitions (i.e. groups of resources associated with specific activities and their working schedules, such as doctors, nurses, and radiologists) and cost information, which were then integrated into the simulation parameters.

4.3. Accuracy assessment

A successful application of Process Simulation for process improvement relies on a simulation model that reflects the real process behavior. Conclusions drawn from an unrealistic simulation model lead to process redesigns that may not yield improvements or even may worsen the performances.

The accuracy of a process model $\mathcal{N}$ is usually assessed using conformance checking techniques [6]. The goal is to identify and analyze deviations between the actual execution of the process (i.e. the recorded data) and the expected behavior as described by the model. Determining the quality of a process model relies on finding the best trade-off between four key measures: *fitness*, *precision*, *generalization* and *simplicity*. Fitness expresses how well the recorded events in an event log align with the process model. A model with high fitness indicates that most of the observed traces can be replayed by the model. Precision evaluates how much permissive is the process model. A model with a low precision is too permissive, allowing for process instances not observed in reality. A model with a perfect precision only allows traces seen in the event log. Generalization assess whether the model can generalize from the observed behavior. A model with a good generalization does not overfit the event log. Simplicity measures the complexity and understandability of the process model.

Assessing the accuracy of a process model is a key step. Chapela-Campa et al. [26] proposed a set of metrics with the scope to measure distances between the original event log and one generated by a process simulator. In our approach, we propose to use these measures, additionally to the classic conformance checking ones (cf. Section 4.2), to assess the quality of the discovered process simulation model. These distances reflect different perspectives, showing discrepancies in the simulator models to be repaired. Accuracy is measured with two control flow, three temporal, and two congestion measures [26]:

- **Control-Flow Log Distance** (CFLD) It computes the average distance to transform each case on an event log (the simulated one) to another case in the another event log (the actual).
- **N-Gram Distance** (NGD) Given two event logs, it measures the “effort” to balance the distributions of the n-grams of the sequences of activities observed in the two event logs. Note that the Earth Mover Distance between the 2-gram histograms is equivalent to the measure proposed by Leemans et al. [32].
- **Absolute Event Distribution** (AED) The event logs are decomposed to build two time series regarding how many events are executed per timestamp, according to a granularity parameter. Then it measures the number of events for balancing the two time series.
- **Circadian Event Distribution** (CED) The two event logs are partitioned into sub-logs by the day of the week, and then Earth Mover Distance between the timestamps distributions is evaluated, measuring the number of events to “move” for balancing the histograms of the frequency of events for each weekday of the two event logs.
- **Relative Event Distribution** (RED) It evaluates how accurate the generation of the timestamp of a subsequent event is based on preceding events. In particular, it measures the “effort” at balancing the distributions of the execution times obtained from the original and the simulated event logs.
- **Case Arrival Rate** (CAR) It works in a similar way as AED, where the time series consider only the arrival events.
- **Cycle Time Distribution** (CTD) It measures the Earth Mover Distance between the distributions of the cycle time of the traces between the two event logs.

These measures quantify discrepancies between the original and simulated event logs, providing information on potential improvements for the simulator. For formal definitions and detailed formulations of these metrics, we refer the reader to the work of Chapela-Campa et al. [26]. If the simulation model generates event logs close to the actual one, according to these metrics, it can be considered accurate. Consequently, it can be used for conducting *what-if* analysis and optimizations.

In this paper, we use normalized metrics, obtained by dividing each value by the maximum possible distance, resulting in values ranging from 0 to 1. This enables an easier interpretation of model quality: values near 0 indicate better similarities between the simulated and actual processes, while values that become closer to 1 suggest a poorer similarity.

4.4. Process improvement

Process Simulation techniques enable the conduction of *what-if* experiments, avoiding the need for building or disrupting the real-world system, with the scope to analyze and improve processes. By examining various *what-if* scenarios, it is possible to assess the impact of potential changes on the processes and the associated protocols and norms that define the current process.

In Section 2.2, we discussed how an optimization problem can be defined to identify configurations that optimize a specific KPI through simulation. In the healthcare domain, a main goal is to improve patient journeys while at the same time ensuring that the overall process remains sustainable. An optimal resource allocation may improve the process in several dimensions, for example, reducing patient waiting times, resources overloading, and overall costs. However, some objectives are often conflicting and achieving an effective balance between them is crucial. For example, while reducing patient waiting times can be accomplished by allocating additional resources and/or expanding physical spaces, such improvements inevitably lead to increased operational costs. However, these costs must align with available budgets and policies to ensure the sustainability of the process.

The goal is to improve the balance of multiple KPIs, each modeled as a function $f_i(\mathcal{N}, \mathcal{P})$ that are computed on the behaviors simulated by a given simulation model $\mathcal{M} = (\mathcal{N}, \mathcal{P})$, which of course needs to be as accurate a representation of reality as possible.

In particular, the optimization focuses on determining an optimal resource allocation $\mathcal{P}^{res} \subset \mathcal{P}$:

$$\begin{aligned} \min_{\mathcal{P}^{res} \subset \mathcal{P}} \quad & \mathbb{E}\left[\sum_{i=1}^{n} w_i \cdot f_i(\mathcal{N}, \mathcal{P})\right] \\ \text{s.t.} \quad & g(\mathcal{N}, \mathcal{P}) \leq C \end{aligned} \tag{1}$$

where $w_1, \dots, w_n$ are constant weights, and the constraint $g(\mathcal{N}, \mathcal{P}) \leq C \in \mathbb{R}^K$, ensures that certain limitations are respected, such as the maximum number of resources or budget constraints.

Weights $w_1, \dots, w_n$ are set to normalize the KPI values and/or to assign larger or smaller importance to some KPIs over others. In fact, Section 5.3.1 illustrates how the weights have been determined for the case study presented in this paper, so as to normalize the influence of the two KPIs involved.

Note that although optimization targets only resource parameters $\mathcal{P}^{res} \subset \mathcal{P}$, the KPI functions $f_i(\mathcal{N}, \mathcal{P})$ account for all simulation parameters $\mathcal{P}$. In fact, some KPIs, such as waiting time, are influenced not only by resource parameters but also by others, such as activity execution times: shorter activity durations can free resources earlier, reducing waiting times. This distinction ensures that we optimize only the resource-related aspects of the process without ignoring the broader dependencies captured by the simulation.

To solve this optimization problem, we propose a simulation-based algorithm that aims to identify the best trade-off solution. The algorithm begins with an initial resource configuration given by $\mathcal{P}^{res}$. Note that this initial configuration could represent a resource allocation where each role has exactly one resource, or it could be an existing configuration from the current system. For each role $R \subset \mathcal{R}$, i.e. a set of resources assigned to a specific role, the algorithm computes the objective function filtering on the role R. Specifically, these values can be computed from a simulated event log $\mathcal{L}^{sim}$, generated using the simulation model $\mathcal{M}$, filtering on the events associated with that role. These values reveal the roles where bottlenecks are most likely to occur. The algorithm then selects the role with the highest value and assigns one additional resource to it, ensuring that all constraints are satisfied. This approach is based on the hypothesis that allocating additional resources to the most critical role effectively mitigates current capacity constraints, thereby reducing bottlenecks and enhancing overall process performance. The procedure is then iteratively repeated until a certain stopping criterion, e.g. when the objective function does not improve. Finally, the algorithm returns a potential optimal solution, representing a trade-off between the KPIs.

The robustness of the optimal configuration can be evaluated by experimenting with different scenarios, such as an increased patient flow. Further analysis can be conducted on various patient attributes to show the efficiency of the resource configuration across diverse patient profiles.

Since existing simulators did not support the methodology discussed here, we implemented our own process simulator in Python, which can be downloaded at https://github.com/franvinci/PetriNetBPS . The simulator builds upon the `pm4py` library [33], supporting the automatic discovery of simulation parameters (cf. Section 4.2).

Our simulator uses Petri net as modeling language [34]; however, the BPMN process models that are used in this paper can be easily converted into Petri nets [35].

5. Case study

This section presents the evaluation of our method through a new real-life case study concerning the process of an emergency department of an Italian hospital. The case study is aimed to improve the process at the emergency department and fairly minimize the waiting time of patients in their journeys, while keeping the process sustainable in terms of costs. In particular, the waiting time for a patient is here considered as the sum of waiting times for each activity carried out at the emergency department. The waiting time for an activity for a patient is the delay between when the activity is ready to be performed for the patient in question and when the activity is actually started. This delay is typically caused by capacity constraints (e.g., resources availability), which prevent an immediate start of the activity. In this urgency case scenario, waiting time represents a critical key performance indicator: lower waiting time reduces the patient risks, and increases patient satisfaction because they are treated more promptly. At the same time, managing operational costs related to resource usage is essential for maintaining a sustainable and efficient healthcare service.

We first introduce the case study, describing the dataset and the process (cf. Section 5.1). In Section 5.2 we discuss the modeling and the accuracy evaluation steps. In Section 5.3 we applied our methodology to derive an optimal resource configuration, further evaluating its robustness in variable patient flow scenarios. Finally, Section 5.4 reports on the discussion of the results with the emergency department staff.

5.1. Introduction to the event dataset and process

The construction of the process simulation model for process improvement has been based on an event log that comprises 37 942 traces with a total of 368 081 events, covering eight months of process data. Each trace corresponds to the journey of a patient through the Emergency Department, from triage to discharge. The original data used in this study cannot be shared due to confidentiality. Table 1 illustrates the excerpt of the event log, which covers the events observed for two traces, namely for two patient journeys. Table 3 provides some

Table 3
Emergency department event description. First column represents an unique attribute. The second indicates the number of classes of the associated attribute present in the event log. Last column describes the instance attribute.

Attribute	N. Classes	Description
Case Id.	37 942	The trace identifier.
Activity	42	Activity label.
Timestamp	–	Associated timestamp (It could be missing).
Lifecycle	2	Associated lifecycle (*start* or *complete*).
Age	–	Integer representing the age of the patient.
Access type	10	The access type (*Autonomous*, *Ambulance without* or *with doctor*, *Helicopter*, etc.).
Urgency code	5	A color referring to the urgency case (*White*, *Blue*, *Green*, *Yellow*, *Red*).
Pathology	40	Pathology assigned at the triage phase.

Table 4
Average, Standard deviation and median cycle time, expressed in minutes, of a process instance per triage color. Last row shows the statics computed in the full event log, i.e. not filtered by triage color.

Color group	Description	Cycle time (min)		
		Avg	Std	Median
White - 1	Non-Urgent	191.42	1017.72	104.73
Blue - 2	Standard	255.26	3042.29	121.41
Green - 3	Urgent	355.22	3210.74	145.63
Yellow - 4	Very-Urgent	833.08	5272.11	223.93
Red - 5	Immediate	1215.68	7979.34	162.21
Complete		486.27	3986.25	159.38

statistics and information about the event log and the process. The event log includes information about the process-instance identifier, executed activities with associated timestamps and life-cycles, the age of the patient, the access type, an urgency code (triage color), and the pathology assigned at the triage phase.

According to the regional regulations, five triage colors are introduced [36]:

1. *White code*: non-urgent patients whose conditions are not true accidents or emergencies.
2. *Blue code*: standard patients without danger or distress.
3. *Green code*: urgent patients with serious problems, but apparently stable condition.
4. *Yellow code*: very-urgent patients with serious illness or injured patients whose lives are not in immediate danger.
5. *Red code*: immediate patients in need of immediate treatment for the preservation of life.

Each group can have different treatments and priorities, and patient flow may change accordingly to the emergency case. Table 4 reports on the cycle times statistics computed for each triage color, comparing them with those of the complete event log. A notable observation is that an increasing level of urgency leads to higher cycle time and higher standard deviation, due to the variations of more urgent cases. Also, median cycle times differ significantly from the mean, suggesting the presence of several outliers in the case study.

Many events related to the starting of activities are missing. As mentioned, this would make it impossible to compute the activity durations. Therefore, as discussed in Section 4.1, we preprocessed the event log to estimate the timestamps of the missing events related to the starting of activities. As mentioned, the technique to estimate the start-event timestamp is based on process simulation, thus requires a full-fledged simulation model, except for the activity duration distribution.

The process model was initially discovered via Inductive Miner [37], and later we show it to the process experts, who were largely satisfied but asked to apply some changes. These included adjusting the concurrency block after the *Triage* activity to allow the *Visit* to be executed in parallel with other activities, and introducing a loop for the *Lab Service* activity following *Blood Sampling*. Fig. 3 shows the model using the BPMN notation: in fact, the model is fairly simple and understandable.

Table 5
Frequencies of events and traces for train and test event logs per triage color. The column total is the sum of the frequencies of the train and test logs. Last row shows the frequencies in the full event log, i.e. not filtered by triage color.

Color group	Number of events			Number of traces		
	Train	Test	Total	Train	Test	Total
White - 1	2372	588	2960	384	96	480
Blue - 2	31 685	8482	40 167	4436	1220	5656
Green - 3	148 054	36 600	184 654	16 454	4122	20 576
Yellow - 4	105 390	25 377	130 767	8529	2045	10 574
Red - 5	7971	1562	9533	550	106	656
Complete	295 472	72 609	368 081	30 353	7589	37 942

5.2. Process simulator modeling & accuracy assessment

To assess the accuracy of the process model and the simulation model, we temporally divided the event log into a train and a test set. The first 80% of the traces in chronological order were allocated to the training set, while the last 20% to the test set, following a commonly adopted approach when assessing the accuracy of simulation models [10,11,38]. Table 5 shows the frequencies of events and traces in both sets, also filtering them by urgency code. The training set is used to build the process model and simulation model. The test set is employed to compute the metrics defined in Section 4.3, which aim to evaluate the accuracy of the discovered simulation model.

Section 5.1 concluded with the discussion of the discovery of the process model in Fig. 3. This process model is characterized by fitness of 0.97 and precision of 0.82, which were both computed on the test sets. The model also exhibits a high level of generalization with a score of 0.93, highlighting that it is not overly restricted to the behavior observed in the event log. The high values for all of the three metrics indicate that the process model is capable to faithfully represent the process' behavior. It is therefore an excellent candidate for use in simulation.

For the construction of the simulation model, the parameters were trained as discussed in Section 4.2, except for the resources, roles, and their calendars, which were defined separately. Unfortunately, the event log did not contain resource information for role and calendar discovery. The lack of resource-perspective information was compensated by inspecting the hospital information and interviewing process experts: the Emergency Department operates 24 h a day, 7 days a week. Each role is associated with specific activities. Table 6 provides a detailed overview of the number of resources available per hour for each role, the associated activities and the cost per hour, which is useful for further analysis.

The fully-fledged simulation model was then used ten times to generate ten simulated event logs, each containing the same number of traces as the test event log. If the simulation model is accurate, the simulated and test event logs are sampled from the same distribution. The metrics introduced in Section 4.3 are indeed meant to assess this. These metrics are then averaged across the ten simulated event logs, thus mitigating the intrinsic stochasticity of running a simulation.

These averages are reported in Table 7, and they are also filtered by different triage colors. Recall that these metrics are normalized in the range $[0, 1]$, where zero indicates a perfect match of the distribution between the simulated and test event logs.

CFLD, 2GD, and 3GD metrics generally score relatively low and similarly on average, particularly for the most common colors given at triage, namely white, blue, and green, which together account for 70% of the cases (26,711 out of 37,942 traces, as shown in Table

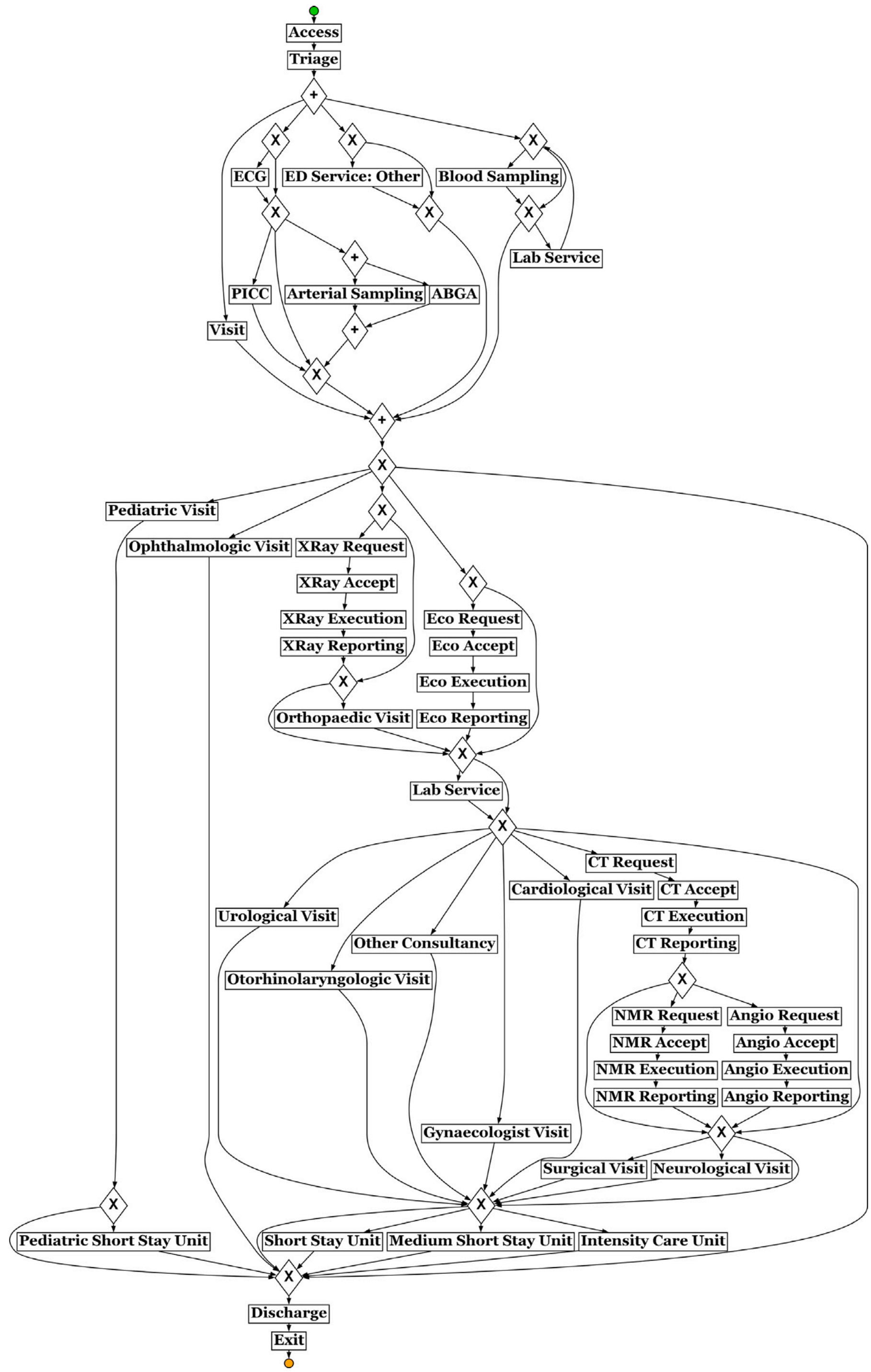

Fig. 3. Emergency department process model in BPMN notation. The green circle represents the initial state, while the orange represents the final. Labels in the rectangles indicates activity tasks. "+" and "X" represent respectively AND and XOR splits/joins. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5). This is largely because the discovery of the simulation model is naturally aiming to represent the most frequent case types. Notably, 3GD, which measures similarity across sequences of three events, tends to yield worse scores than 2GD and CFLD: capturing longer activity dependencies is inherently more complex. The model performs less effectively for traces associated with a red color at triage, especially in terms of the 3GD metric. This is likely due to a significant imbalance in the training data, where red-colored traces make up only 1.8% (550 out of 30,353) of the total training traces. Nonetheless, these instances are rare and have limited impact on the overall performance of the simulation model. As illustrated in Table 7, the weighted average remains largely unaffected by the lower scores associated with executions for patients who are assigned a red color at triage. In conclusion, the simulation model can well mimic the control-flow perspective of the emergency-department healthcare management process.

The other metrics are related to the temporal perspectives and assume very low values: this highlights that they are well simulated by the model. The temporal perspective is also tightly connected to the resource perspective: if the resources, with their schedules and assignment to roles and activities, were poorly modeled, this would

Table 6
Number of resources per role with their associated activities and cost per hour.

Role	Resources per hour	Cost/h	Activities
Doctors	4	75€	*Visit, Discharge, ED Service Other*
Nurses	5	30€	*Blood Sampling, ECG, PICC, Arterial Sampling, ABGA*
Triage N.	2	30€	*Triage*
Radiologist	1	75€	*XRay, Eco, CT, NMR, Angio Reporting, Eco Execution*
Rad. Tech.	2	30€	*XRay, CT, NMR, Angio Execution,*
Beds	12	30€	*(Pediatric) (Medium) Short Stay Unit, Intensity Care Unit*
Consultant	1 per kind	75€	*Pediatric, Cardiological, Orthopaedic, Surgical, Neurological, Urological, Ophthalmologic, Gynaecologist, Otorhinolaryngologic Visit*

Table 7
Simulation distance measures computed averaging the results obtained across various simulations. The rows with triage colors show results obtained by filtering the test event log and the simulated event logs by the corresponding color. Recall values are normalized between 0 and 1, where 0 indicates no distance.

Event log	CFLD	2GD	3GD	AED	CED	RED	CAR	CTD
Complete	0.13	0.19	0.25	0.08	0.07	0.01	0.09	0.01
White - 1	0.09	0.15	0.18	0.17	0.15	0.05	0.15	0.10
Blue - 2	0.07	0.16	0.19	0.08	0.08	0.01	0.09	0.01
Green - 3	0.11	0.17	0.22	0.09	0.07	0.01	0.09	0.01
Yellow - 4	0.20	0.26	0.35	0.07	0.07	0.01	0.08	0.01
Red - 5	0.30	0.36	0.47	0.14	0.08	0.01	0.13	0.02

affect the realism of the waiting queues, namely the queues of activities that are waiting for starting because all resources are not available at that point. Poor waiting-queue realism would consequently degrade the realism of the temporal perspective: for instance, the simulation's waiting times would not match the real waiting times.

In summary, the simulation model shows a high level of realism for most of potential behavior, and thus can be used to improve the process, aiming to improve on the resources allocation.

5.3. Process improvement

Once the simulator is built, it can be used for improvement purposes as described in Section 4.4. In the emergency department case study, one of the main objectives is to identify an optimal resource allocation that minimizes patient waiting times while ensuring the process cost remains sustainable. Section 5.3.1 presents the optimization results. Section 5.3.2 discusses a robustness analysis in cases of varying patient flows, i.e. the number of patients arriving at the hospital.

5.3.1. Resource allocation improvement

The case study goal is to minimize the expected **waiting time of patients** into the hospital, thus minimizing the overall times and improving their journeys. The expected patient waiting time can be computed via simulations. Given a simulated event log $\mathcal{L}^{sim}$, obtained from a simulation model $(\mathcal{N}, \mathcal{P})$, with $\mathcal{P}^{res} \subset \mathcal{P}$ the set of resource parameters (cf. Section 4.4), the expected waiting time is computed as follows:

$$\mathbb{E}\left[f_1(\mathcal{N}, \mathcal{P})\right] = \frac{1}{|\mathcal{L}^{sim}|} \sum_{\sigma \in \mathcal{L}^{sim}} \sum_{e \in \sigma} wt(e) \tag{2}$$

where $wt(e)$ indicates the waiting time of the activity instance $act(e)$ to which $act(e)$ refers, which is here computed as the difference between the time in which $act(e)$ starts and the time in which the activity instance preceding $act(e)$ completes. The rationale is that high waiting times are due to the congestion of some resources, then increasing the number of the resources related to the distinguish roles could decrease the waiting times associated with the activities executed by them. However, not adequately increasing the number of resources can lead to unsustainable costs. Then, the **total resource cost** must also be managed during the optimization process. The goal is hence to find a good trade-off between total cost and expected waiting times. The total hourly cost can be computed as follows:

$$f_2(\mathcal{N}, \mathcal{P}) = \sum_{r \in \mathcal{R}} Cost(r) \tag{3}$$

where $\mathcal{R}, Cost \in \mathcal{P}^{res}$ are the parameters referring to the set of available resources $\mathcal{R}$, and the hourly cost function $Cost$ returning the cost per hour of a certain resource. Note that since $f_2(\mathcal{N}, \mathcal{P})$ is constant for fixed $(\mathcal{N}, \mathcal{P})$, it follows that $\mathbb{E}[f_2(\mathcal{N}, \mathcal{P})] = f_2(\mathcal{N}, \mathcal{P})$.

The optimization problem in Eq. (1) is defined here using the two KPI functions defined in Eqs. (2) and (3). Specifically, note that:

$$\begin{aligned}\mathbb{E}\left[w_1 \cdot f_1(\mathcal{N}, \mathcal{P}) + w_2 \cdot f_2(\mathcal{N}, \mathcal{P})\right] &= w_1 \cdot \mathbb{E}\left[f_1(\mathcal{N}, \mathcal{P})\right] \\ &\quad + w_2 \cdot \mathbb{E}\left[f_2(\mathcal{N}, \mathcal{P})\right] \\ &= w_1 \cdot \mathbb{E}\left[f_1(\mathcal{N}, \mathcal{P})\right] + w_2 \cdot f_2(\mathcal{N}, \mathcal{P})\end{aligned} \tag{4}$$

where the weights are selected to normalize and assign equal importance to both KPIs, f_1 and f_2. Specifically, w_1 and w_2 are determined based on the maximum possible values that their corresponding KPI functions can achieve. The weight w_1 is set to $1/45.25$, where 45.25 min represents the current average waiting time. The weight w_2 is set to $1/(1830 + N \cdot 75)$, where 1830€ is the current hourly cost, and the term $N \cdot 75$ accounts for the potential increase in cost after adding N resources—each contributing to a maximum of 75€ per hour (cf. Table 6). This ensures that both performance measures are scaled comparably, allowing for balanced optimization.

For this case study, we defined the constraint where resources from the current configuration must be retained. Removing resources is not permitted, as it could lead to unfair conclusions where certain resources might be laid-off. To satisfy this, we initiated the algorithm by using the current configuration (see Table 6), thus ensuring that its resources cannot be removed. The optimization problem is then solved by using the algorithm proposed in Section 4.4.

We experimented with a maximum of 10 iterations. The most substantial improvements occur within the first two iterations, after which the optimization process stabilizes. Fig. 4 illustrates the average waiting time (y-axis) and total hour cost (x-axis) for each configuration, highlighting the rapid initial gains followed by a plateau in performance. The optimal solution, i.e. the one minimizing the objective function, is highlighted in green. Specifically, it chooses as the best configuration the one obtained after two iterations, leading to a configuration with *one more radiologist and one more orthopedic*, which is expected to decrease the waiting time of ca. 89%, in exchange for a modest hourly-cost increase of ca. 8%.

5.3.2. Readiness analysis to sudden workload peaks

Once the best configuration was identified, we focused on the ability of the system to tackle sudden peaks in the arrival rate, e.g. due to an emergency scenario. Section 1 discussed the importance of healthcare systems to respond to sudden peaks. This readiness analysis adopted the methodology from [22] by adjusting the patient flow. Specifically, we tested arrival rates with a patient-flow ratio set at 0.5, 1.0, 1.5 and 2.0 times the baseline arrival rate of the original configuration. For example, a patient ratio of 0.5 corresponds to a scenario in which the patient flow decreases of 50%, 1 to the current patient flow, 1.5 signifies a 50% increase and 2 a scenario in which patient arrivals double.

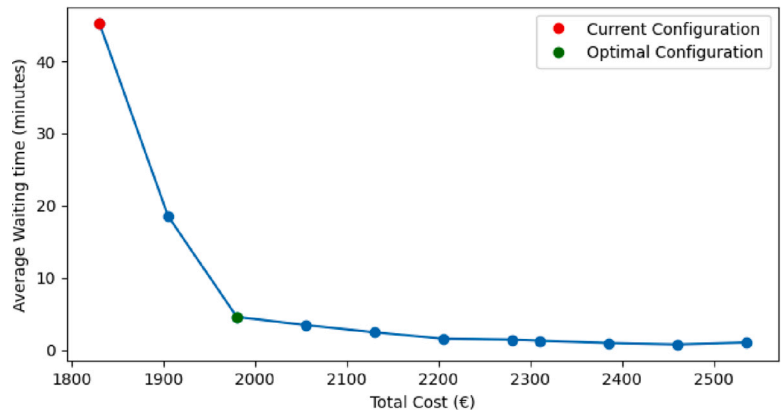

Fig. 4. Average waiting time (in minutes) per total cost per hour (in €). Each point represent a resource configuration. The red point represents the results using the initial, i.e. the current, resource allocation. The green point indicates the optimal solution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

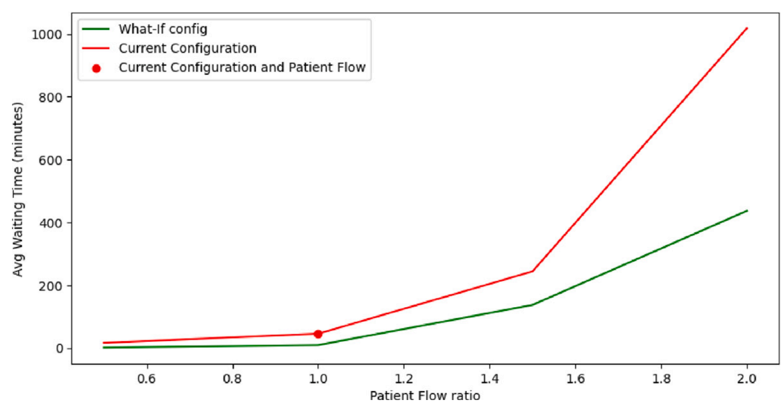

Fig. 5. Average waiting time (in minutes) per patient flow ratio. Results are obtained averaging values from various simulations. The red line represent results obtained using the current configuration with various patient flow ratio. The green highlights those obtained by using the optimal configuration identified by our method. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For each patient ratio, we simulated 10 event logs and computed the average of the waiting times across these simulations. The results are depicted in Fig. 5, and reveal a better readiness of the improved process configuration to emergency scenarios, compared to the initial process configuration: waiting times exhibit a significantly faster increase for the initial process configuration. This highlights the efficacy of our optimization approach in managing workflow efficiency and mitigating delays, thereby ensuring more stable and predictable performance under varying patient ratios.

5.4. Discussion with ED staff: Model check and operational insights

The simulation model and the corresponding results were presented to ED staff in a dedicated debriefing session. The discussion served two purposes: (i) a plausibility check of model structure and performance realism against day-to-day operations, and (ii) interpretation of scenario results to surface pragmatic improvement actions. Overall, ED clinicians confirmed that the control-flow, resource roles, and performance levels reflected by the model align with how the ED actually works. Specifically, the set of activities and routing logic mirrored their experience of parallel diagnostic and/or treatment paths following triage and visit. This is consistent with the process model previously discovered, which they recognized as a faithful abstraction of their workflows.

A key point of convergence between the simulated evidence and staff perception concerned imaging. Teams emphasized that radiology behaves as a heavily utilized single “server” that periodically attracts a surge of simultaneous requests from multiple pathways (e.g., imaging after the initial visit, or in parallel with blood sampling/lab work), creating queue build-ups during peak windows. This observation is aligned with the model structure and with the resource distribution (cf. Table 6), which shows limited radiological reporting/execution capacity relative to the number of upstream activities that can generate demand. In practice, this concurrency produces short periods of high contention even if average loads look manageable, explaining why Radiology appears as the dominant bottleneck in both data and practice. Conversely, the team found the result concerning the orthopedic specialist less intuitive. While an orthopedist dedicated to the ED is clinically valuable, they noted that orthopedic consults do not arrive at a steady pace; instead, quiet periods alternate with bursts. Under these conditions, a single, fixed consult channel can intermittently inflate waits for the subset of patients requiring orthopedics, even when other resources are momentarily available. This observation does not contradict the *what-if* analyses that identified an additional orthopedist (together with a radiologist) as part of a cost-effective configuration overall; rather, it highlights that consult scheduling and flexibility matter when arrivals are bursty. Staff also recalled that, in prior years, orthopedists remained on their ward and were called in when needed, rather than being assigned as a dedicated ED resource. In that time, they had longer average waiting times for patients, but they were a little more flexible when they had demand peaks.

The staff’s appraisal of improvement options was generally positive. There was strong consensus that increasing radiology capacity dedicated to ED demand, for example, via additional radiologist coverage during peak hours (also shared by other EDs since the reporting is possible “online”) and/or an extra technician on high-demand shifts, is a sensible and immediately actionable lever. This judgment coheres with the *what-if* evidence showing that targeted additions on the imaging side yield outsized reductions in average waiting time at modest marginal cost. Although the exact configuration will depend on numerous factors, the direction of change was considered realistic and operationally acceptable by the ED leadership. At the same time, rather than simply “adding” more dedicated orthopedic coverage, the discussion suggested organizational refinements to absorb burstiness without investing too many resources and creating new rigid bottlenecks: e.g., on-call orthopedist responding within set time windows when the ED consult queue exceeds a threshold or pooling ED orthopedic with the ones assigned to the hospital ward for cross-cover during peaks.

6. Threats to validity and future directions

While our methodology has shown its value through the case study presented in this work, both the methodology and its empirical evaluation are subject to certain assumptions and limitations that may affect their generalizability and applicability to other healthcare contexts. In this section, we discuss the main methodological and experimental assumptions and limitations, and outline possible avenues for future research.

6.1. Limitations of the methodology

The proposed methodology builds upon traditional process discovery techniques that assume the availability of one representative event log. While our emergency-department case study relied on a single information system, other health-care processes can be distributed over multiple departments, each with a different information system, leading to fragmented and incomplete event data. It may be the case that the scattered event data from different source share the same identifier (e.g. the social security number) or, if they have different identifiers, these can be linked to each other (e.g. the patient identifier in one system and the social security number in other). However, we acknowledge that this is not guaranteed: if the fragmented event data cannot be merged, we cannot have a single representative event log, and thus our methodology, and more in general Process Mining, cannot be applied.

The methodology is particularly effective for well-structured healthcare processes, where workflows follow standardized procedures, resources can be allocated in advance, and process logic can be accurately captured in an event log. Conversely, highly flexible or knowledge-intensive processes, such as those involving personalized patient care pathways or complex diagnostic coordination, may not be fully captured by our approach. As illustrated in the emergency department case study, our methodology is best suited for **organizational healthcare processes**, following the categorization proposed by Lenz et al. [7], while its applicability may be more limited in contexts characterized by high variability, uncertainty, or extensive clinical decision-making.

In representing temporal aspects, we assume that delays are only caused by resource capacity constraints. This tends to be true for processes that time critical, such as in hospital’s emergency departments, which is indeed the domain of the case study in Section 5. As a matter of fact, Table 7 reports that the simulated event log shows a distance equal to 0.07, with respect to Cycle Time Distribution (CTD), where distances are normalized between 0 and 1. Since CTD is related to the temporal properties of the (simulated) process, this illustrates that the assumption holds for the case study in question. We however acknowledge that this abstraction does not explicitly model intricate temporal dependencies or synchronization constraints that may occur in real-world healthcare workflows [39], such as minimum or maximum time distances between activities [40] or externally induced delays [41]. Consequently, for processes where inter-activity timing and coordination play a critical role, more advanced temporal modeling techniques would be needed to accurately capture process dynamics.

Moreover, our methodology primarily focuses on the organizational dimension of healthcare process improvement: resource allocation, cost, and time. In the specific case study discussed Section 5, a modest increase in personnel cost of ca. 8% leads to the patient’s waiting time to reduce to 11% of the original situation. However, Challenge C8 [15] also indicates that process optimization should also be viewed “through the patient’s eyes”. This includes minimizing unnecessary diagnostic procedures, ensuring that invasive examinations are performed only when non-invasive ones are inconclusive, and designing diagnostic and therapeutic pathways that reduce patient burden. Addressing such aspects would require extending the health-care methodology in Table 2 to allow altering the control-flow model during the methodology, along with increasing simulation-model expressiveness to explicitly encode clinical constraints and dependencies among diagnostic and therapeutic procedures

Finally, the proposed methodology does not account for **concept drift** or process seasonality, which are common in real-world healthcare settings (cf. Challenges 3 and 10 in [15]). For instance, patient arrivals may significantly increase during specific times of the year (e.g., flu season) or fluctuate across different hours of the day. These temporal variations can affect the reliability of the optimization if not properly considered.

6.2. Limitations related to the case study

We acknowledge that the case study presented in Section 5 refers to a single department, namely the emergency department, and does not contain more intricate coordination and temporal dependencies, such as those present in the case study presented in [42]. However, as discussed in Section 6.1, if a complete event log could be constructed by integrating traces from multiple departments, every indication suggests our approach could still be applicable, while we would need to assess the capability to model the specific intricate coordination and temporal dependencies, if they were present. Therefore, this limitation largely reflects constraints in data availability rather than a deficiency of the methodology itself.

In addition, as is common in healthcare process data (cf. Section 7), the event log also contains missing information regarding activity starts and incomplete resource attributes. These gaps were addressed by estimating missing timestamps and retrieving resource information through consultation with stakeholders. However, this procedure may introduce minor inaccuracies and, since resource data are not linked at the event level, detailed analyses could be affected.

Furthermore, the evaluation of the simulation model and its results relied on interpretative feedback gathered through discussions with ED personnel rather than on a structured approach. While this participatory approach provides valuable contextual insights and fosters practical relevance, it may limit the reproducibility of the results.

Finally, the proposed improvements have not been implemented or evaluated in real-life environments. While the data-driven simulation framework allows the process to be modeled and analyzed in an objective, evidence-based manner, all optimizations reported in this study are based on simulated scenarios rather than real-life execution.

6.3. Future work

In light of the aforementioned limitations, future research will aim to extend the proposed methodology to better capture the temporal and behavioral complexity inherent in many healthcare processes. As discussed, the current framework primarily explains activity durations and waiting times through resource availability. However, in real clinical environments, additional sources of delay often play a significant role.

Incorporating such external delay factors, as addressed by Chapela-Campa et al. [41], could improve the realism and predictive accuracy of the simulation framework.

Moreover, we currently build on imperative process modeling notation (cf. Petri net and BPMN models). Their structuredness might not be satisfactory to model the flexibility that is typical of healthcare processes. To enable flexible control-flow logic, we plan to integrate declarative process modeling into the simulation framework. Declarative models, such as those expressed in the Declare language, define behavioral constraints without enforcing strict execution orders, making them suitable for dynamic and knowledge-intensive healthcare processes [42–45]. Advances in declarative process simulation [46] and multi-perspective model discovery [47] provide promising foundations for capturing variability and clinical decision interdependencies more effectively.

An additional promising direction for future research concerns the management of concept drift in healthcare processes. Clinical workflows naturally evolve as patient populations, medical protocols, and resource conditions change, potentially affecting the stability of discovered models. Recent research [48] proposes techniques for adapting to drift in process execution data, ensuring models remain accurate as conditions evolve. Building on this approach, future extensions of our methodology could incorporate drift-aware simulation mechanisms that dynamically update model parameters, thus preserving accuracy and reliability in continuously evolving healthcare environments.

Finally, future research should focus on a systematic approach to evaluate the proposed simulation model and its results through the structured involvement of healthcare professionals. This evaluation could employ systematic approaches such as focus groups and semi-structured interviews to ensure reproducibility. Moreover, additional optimization strategies identified by ED staff could be experimentally evaluated within the simulation environment. Such additional testing would yield stronger empirical evidence of the model's practical utility and its effectiveness in supporting data-driven decision-making and operational optimization in complex healthcare systems.

7. Related works

The healthcare sector increasingly leverages data-driven methodologies to gain insights into complex processes and identify opportunities for improvement. In this section, we first review the application of Process Mining techniques for analyzing and understanding healthcare workflows using event data (cf. Section 7.1). Next, we discuss the literature on process simulation approaches for modeling, evaluating, and improving healthcare processes (cf. Section 7.2).

7.1. Process mining in healthcare

Process execution data stored in Healthcare Information Systems can be used through Process Mining techniques to support the management and optimization of healthcare processes. Extensive research has explored the use of Data and Process Mining techniques for analyzing these processes using event data, revealing distinguishing characteristics and associated challenges to be addressed [15,49–54].

A significant part of works focused on analyzing and modeling processes to draw conclusions and provide recommendations for improvement. Augusto et al. [27] used process data to analyze the impact of the COVID-19 pandemic in an Australian hospital. Mans et al. [55] used Process Mining to gain insight into the care flow of a Dutch hospital from three different perspectives. Leemans et al. [56] introduced a novel model incorporating process-based micro-costing estimations, which can be used in a decision analytics context. In [57] the authors proposed a method to model surgical processes that involve experts in the healthcare domain. Rabbi et al. [58] analyzed an emergency department process using Process Mining algorithms.

Interpretability and understandability pose significant challenges in the healthcare sector [53,59]. Although Process Mining techniques are recognized as white-box, many models remain complex and difficult to interpret. Other works focused on enhancing the understandability of process models in the Healthcare domain. Li et al. [60] proposed a novel process model miner to construct interpretable process models for complex medical processes. Furthermore, [61] demonstrated the benefits of the interaction of a domain expert in improving the quality and understandability of process models.

Healthcare processes tend to exhibit substantial variability [15], for example, certain process paths can be taken only by some patients with specific characteristics. Considering the variability of process actors is essential for effective process management. Andrews et al. [62] developed an approach to detect process variations between various groups of patients. In [29] the authors introduced a Process Mining methodology to explore the disease-specific care process. Mazhar et al. [63] proposed an automatic method incorporating the stochastic perspective to determine the likelihood of a particular pathway for certain patient cohorts. Bakhshi et al. [64] applied Process Mining algorithms to find decision rules based on features available in the event logs, revealing patient trajectories patterns.

Another addressed key challenge is to deal with low quality data. Real life data from Healthcare Information System often lack precision and are improperly collected, resulting in inaccurate data analysis and process modeling leading to not effective recommendations and optimizations. Several studies proposed alternatives for addressing these issues in the Healthcare domain. Vanbrabant et al. [65] proposed a framework to categorize possible data quality problems in emergency department electronic healthcare records. In [66], 27 possible classes of quality issues are subdivided into four main categories: missing, incorrect, imprecise, irrelevant data. Suriadi et al. [67] document a collection of typical quality problems in event logs; Leonardi et al. [68] introduced an explainable methodology using Deep Learning models for process trace quality assessment.

Van Hulzen et al. [69] analyzed the effects of data quality issues on simulation outcomes, showing the importance of domain knowledge in addressing this challenge using a case study in the radiology department of a hospital. Benevento et al. [28] validated the suitability of Interactive Process Discovery in handling low quality data with a real dataset from an Italian hospital, involving the hospital staff; Martin et al. [70] applied interactive data cleaning in an outpatient clinic's appointment system case study.

Focusing on the temporal perspective, Fischer et al. [71] presented an automated approach to identify and quantify timestamp imperfections, defining 15 related metrics. Dixit et al. [30] proposed an interactive framework to repair ordering imperfections in event logs.

7.2. Healthcare process simulation

Process simulation is a key methodology for monitoring, analyzing and improving healthcare processes. By building data-driven models based on real-world event logs, simulation enables the exploration of different scenarios through *what-if* analyses and supports evidence-based decision-making. Typically, process models are derived using Process Mining techniques, which focus on extracting control-flow, temporal, and resource perspectives from healthcare event data [5,72]. Some works provided automated frameworks for discovering simulation models [10], resulting in accurate models based on process event data, avoiding biases that could arise from relying solely on the subjectivity feedback of the process actors. Additionally, Process Mining techniques have been compared with Deep Learning approaches, showing that these could lead to more accurate simulations, particularly in the temporal perspective [38]. However, although Explainable AI methods can enhance the interpretability of such models, they do not enable direct modification of model behavior, thus limiting their applicability for performing *what-if* analyses [38].

However, Deep Learning techniques are *black-box*, making them unsuitable for certain applications and for performing *what-if* analyses [38].

Several works have focused on developing simulation models specifically for healthcare applications. For instance, Guastalla et al. [16] employed simulation to improve work scheduling in hospital environments, leveraging historical data to enhance operational planning. Van Hulzen et al. [17] developed a process simulation framework to support capacity management decisions within a radiology department, combining data driven model construction with domain expert validation.

Halawa et al. [18] proposed a simulation-based approach to optimize the physical layout and patient flow in a cardiovascular clinic, aiming to improve operational efficiency. Similarly, Tamburis et al. [19] focused on creating simulation models to evaluate performance indicators and benchmark healthcare services.

Antunes et al. [20] integrated simulation into an optimization framework, using it to compare baseline and improved scheduling policies in an Emergency Department setting. Johnson et al. [21] introduced the ClearPath methodology, which incorporates simulation to explore care pathways while documenting data quality aspects throughout the modeling process.

Further approaches have emphasized patient-centered simulation. Kovalchuk et al. [22] developed a conceptual framework combining Process Mining and simulation to model clinical pathways for different groups of patients, thus improving the realism of the simulations. Abohamad et al. [23] applied simulation to identify bottlenecks and explore improvements in emergency department operations.

Mans et al. [24] investigated the impact of IT system introduction on healthcare processes through simulation, providing insights into the technological effects on operational performance. Finally, Ahmed and Alkhamis [25] used simulation modeling based on expert interviews to design optimization strategies for resource allocation in an emergency department context.

8. Conclusion

Optimizing processes is a critical and complex challenge in the healthcare domain. Optimal process configuration can enhance patient journeys, reduce costs, and improve the overall quality of care. Process Mining techniques offer powerful tools for modeling and analyzing real-life processes [6]. The resulting process models can be used for simulation purposes, enabling stakeholders to observe how potential changes might impact performance without the high costs and risks associated with real-world experimentation. Assessing the accuracy of these models is crucial. In fact, findings derived from experiments performed using an inaccurate process model could be implausible and, hence, not acceptable.

In this paper, we proposed a methodology which leverages Process Simulation to support the optimization of healthcare processes. The proposed methodology begins with the application of Process Discovery techniques, which are used to extract a process model from real execution data. This model is then enhanced with various perspective parameters — such as time, cost, and resource usage — to provide a more complete and accurate representation of the actual process. The accuracy of the simulation model is assessed using the metrics proposed by Chapela-Campa et al. [26] which consider the different perspectives. Finally, the simulation model can be used for optimization purposes. We specifically introduce an automated optimization approach that uses simulation to identify optimal process configurations. The goal of this optimization is to balance multiple Key Performance Indicators (KPIs), such as minimizing patient waiting time while controlling or reducing operational costs.

We applied our methodology in an emergency department case study. In this context, we formulated a stochastic optimization problem aimed at identifying a resource configuration that would reduce patient waiting times without leading to unsustainable increases in cost.

Finally, we evaluated the robustness of this optimal configuration through a series of *what-if* scenarios in which we varied patient flow conditions to simulate potential crisis situations. Findings show the optimal configuration of resources maintains more stable performance, if compared with the current one, in critical scenarios, suggesting its effectiveness not only under normal conditions but also in more demanding or uncertain operational contexts.

Reproducibility

GDPR considerations and ethical constraints have prevented sharing the real event log that was used as case study in this article. The real event log would potentially allow disclosing sensitive clinic information about patients that were tried in the hospital considered as case study. Specific pathologies determine specific activity paths, and these paths could be reverse-engineered to infer the underlying condition. Similar reasoning applies to age, timestamps, and other patient-related attributes. A severe form of anonymization could make the re-identification of individual patients practically impossible, thereby ensuring compliance with the GDPR. However, such a dataset would be significantly altered: as a consequence, it would no longer enable reproducibility of the results. To ensure reproducibility while complying with GDPR and ethical constraints, we provide a fully synthetic event log that statistically and structurally resembles the original dataset used in this study. The synthetic log is generated using the 'baseline' technique described in [73] and is publicly available at https://doi.org/10.5281/zenodo.18785676. The similarity between the original and synthetic logs has been quantitatively assessed using the measures proposed by Chapela-Campa et al. [26], including 2GD and 3GD for control flow, AED and RED for event distributions, CAR and CTD for temporal characteristics, and CED for circadian event patterns. In particular, the computed distances, normalized in the range [0,1], are 2GD = 0.0185, 3GD = 0.0333, AED = 0.0431, RED = 0.0003, CAR = 0.0592, CTD = 0.0004, and CED = 0.0858. These values indicate a close correspondence between the original and synthetic logs, confirming that the synthetic data preserve the main behavioral and temporal properties relevant to the simulation experiments. While exact numerical replication of all results cannot be guaranteed due to the synthetic nature of the data, the released log enables independent reproduction of the methodological pipeline and supports transparent validation of the reported findings without exposing sensitive patient information.

Ethics statement

This research project was approved by the University Bioethics Committee of the University of Pisa on 23/04/2025 (Resolution No. 23/2025), which considered the research project to be in line with the principles of free and responsible science, including adequacy with respect to the profiles of data protection: all procedures were performed in compliance with relevant laws and institutional guidelines.

CRediT authorship contribution statement

Francesco Vinci: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Davide Aloini:** Validation, Resources, Investigation. **Elisabetta Benevento:** Validation, Resources, Investigation, Data curation. **Alessandro Stefanini:** Validation, Resources, Investigation, Funding acquisition, Data curation. **Francesca Zen:** Software. **Massimiliano de Leoni:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Assessing the quality of event data: Start timestamp estimation

In this appendix, we present a novel approach for estimating the start timestamps of events in an event log when they are missing. We extend the work of Fracca et al. [31], introducing a new algorithm designed to improve both precision and computational efficiency. In particular, in this paper we propose to search the parameters for estimating the start timestamps using optimization functions depending on the activity leveraging on an algorithm able to reach better local optima.

The general idea is to develop an accurate process simulator that relies exclusively on the information available in the original event log and generates simulated event logs by varying the activity duration parameters. The objective is to find the model that produces runs that are closest to the initial event log. Finally, the parameters employed in the optimal simulation model can be applied to estimate start timestamps within the original event log.

In the following, a first section outlines preliminary concepts relevant for the definition of our method. Appendix A.2 describe the methodology and finally Appendix A.3 reports the evaluation results on two different case studies, comparing them to those obtained using the algorithm proposed in [31].

A.1. Preliminaries

In this section, basic definitions of events, traces, and event logs are presented. Then, temporal perspective definitions are provided to define the method in a following section.

Definition 1 (*Events*)**.** Let $\mathcal{A}$ be a set of activities, $\mathcal{T}$ a set of timestamps, $\mathcal{R}$ a set of resources, $\mathcal{I} = \{start, complete\}$ the set of lifecycles. An event $(a, t, r, i) \in \mathcal{A}\times\mathcal{T}\times\mathcal{R}\times\mathcal{I}$ is a tuple indicating that a resource r performed (the start or the end, according to i, of) an activity a at timestamp t.

In the remainder, given an event $e = (a, t, r, i)$, we define $act(e) = a$, $time(e) = t$, $res(e) = r$, $life(e) = i$.

Definition 2 (*Traces & Event Logs*)**.** Let $\mathcal{E}_{\mathcal{A}}$ be the universe of events defined over a set of activities $\mathcal{A}$. A trace $\sigma = \langle e_1, \dots, e_n\rangle \in \mathcal{E}^*_{\mathcal{A}}$ is a sequence of events such that for all $1 \le i < j \le n$, $time(e_i) \le time(e_j)$. An event log is a set of traces, i.e. $\mathcal{L}_{\mathcal{A}} \subseteq \mathcal{E}^*_{\mathcal{A}}$.

In the remainder, we use the abbreviation $e \in \mathcal{L}_{\mathcal{A}}$ to indicate that there exists a trace in $\mathcal{L}_{\mathcal{A}}$ that contains the event e. Also, we omit the subfix $\mathcal{A}$ from $\mathcal{E}_{\mathcal{A}}$ and $\mathcal{L}_{\mathcal{A}}$ when it is clear from the context.

The following defines the time interval, in terms of temporal units, between two consecutive end events in a trace. This definition will be useful for formulating the objective function to minimize in Appendix A.2.

Definition 3 (*Inter-Event Time*)**.** Let $\mathcal{L} \subseteq \mathcal{E}^*$ be an event log. Let $\overline{\mathcal{L}} = \{\overline{\sigma} \mid \overline{\sigma} = \langle e \mid life(e) = complete\ \forall e \in \sigma\rangle\ \forall \sigma \in \mathcal{L}\}$ be the sub-event log of complete events. We define the inter-event time function $\Delta_{\mathcal{L}} : \overline{\mathcal{L}} \to (\mathcal{E} \to \mathbb{R}^+)$ s.t. $\Delta_{\mathcal{L}}(\sigma) = \delta_\sigma$, where given a trace $\sigma = \langle e_1, \dots, e_n\rangle \in \overline{\mathcal{L}}$, for all $1 \le i \le n$:

$$\delta_\sigma\left(e_i\right) = \begin{cases} 0 & \text{if } i = 1 \\ time(e_i) - time(e_{i-1}) & \text{otherwise} \end{cases} \tag{A.1}$$

where $time(e_i) - time(e_{i-1})$ represents the duration time between event e_i and e_{i-1}.

These durations are used to estimate the following distribution functions for each activity in an event log:

Definition 4 (*Activity Inter-Event Time Distribution*)**.** Let $\mathcal{L}$ be an event log defined over a set of activities $\mathcal{A}$. We define the inter-event time distribution function $\Phi_{\mathcal{L}} : \mathcal{A} \to \left(\mathbb{R}^+ \to [0, 1]\right)$ where $\Phi_{\mathcal{L}}(a)$ indicates distribution of inter-event times $\delta_\sigma(e)$ where $e \in \sigma$ s.t. $act(e) = a$, for all $\sigma \in \mathcal{L}$.

A.2. Method

The starting points of the method are an event log $\mathcal{L}$ defined over a set of activities $\mathcal{A}$, and a simulator model $\mathcal{M} = \left(\mathcal{N}, \mathcal{P}\right)$ where $\mathcal{N}$ is a process model, such as a Petri Net or a BPMN model, and $\mathcal{P}$ is a set of parameters. Since we aim at estimating the event log start timestamps, we assume that $life(e) = complete\ \forall e \in \mathcal{L}$.

To estimate the starting event timestamp of an event $e \in \mathcal{L}$ s.t. $life(e) = complete$ and $act(e) = a$, we formulate the problem as finding a value $\alpha(a) \in [0, 1]$. Particularly, the starting event timestamp $t_{start}(e) \in \mathcal{T}$, where $\mathcal{T}$ is the set of timestamps, can be parameterized as:

$$t_{start}(e) = \alpha(a) \cdot mintime(e) + (1 - \alpha(a)) \cdot time(e) \tag{A.2}$$

where $mintime(e)$ is the minimum timestamp when event e could have started. We estimate the minimum starting timestamp $mintime(e)$ for an event e as follows:

Definition 5 (*Minimum Starting Timestamp*)**.** Let $\mathcal{L}$ be an event log s.t. $life(e) = complete$ for all $e \in \mathcal{L}$. Given a trace $\sigma = \langle e_1, \dots, e_n\rangle \in \mathcal{L}$, the minimum starting timestamp of e_i with $1 \le i \le n$, such that $res(e) = r$ is defined as

$$mintime\left(e_i\right) = \begin{cases} time\left(e^r_{prev}\right) & \text{if } i = 1 \\ max\left(time\left(e_{i-1}\right), time\left(e^r_{prev}\right)\right) & \text{otherwise} \end{cases}$$

where $e^r_{prev} \in \mathcal{L}$ is the previous event in $\mathcal{L}$ performed by the resource r, i.e. $res(e^r_{prev}) = r$, $time(e^r_{prev}) \le time(e_i)$ and $\nexists\ e \in \mathcal{L}$ s.t. $res(e) = r$ and $time(e) > time(e^r_{prev})$.

If such e^r_{prev} does not exist, i.e. resource r has not performed any event previously, we set e^r_{prev} as the first event of $\mathcal{L}$, i.e. $time(e^r_{prev}) \le time(e)\ \forall e \in \mathcal{L}$.

The intuition is to consider both the control flow and the resource perspective. We hypothesize that an event can only start after the previous one has completed and that the resource performing the activity cannot be engaged in multiple activities simultaneously. These assumptions are legit for many processes, particularly in the healthcare domain, where patients and resources (such as doctors and nurses) hardly perform activities in parallel. Fig. A.7 illustrates this hypothesis.

The goal is to determine the optimal set of parameters $\alpha(a)$ for all $a \in \mathcal{A}$. The method is summarized by the schema illustrated in Fig. A.6. It starts with an initial configuration $\alpha = \{\alpha(a) \mid a \in \mathcal{A}\}$ enriching the event log $\mathcal{L}$ with estimated start events, given using Eq. (A.2), resulting in an event log $\mathcal{L}^\alpha$. This is then used to enhance a given process simulator with temporal information (e.g. activity execution time distributions). The simulator is subsequently run to generate a

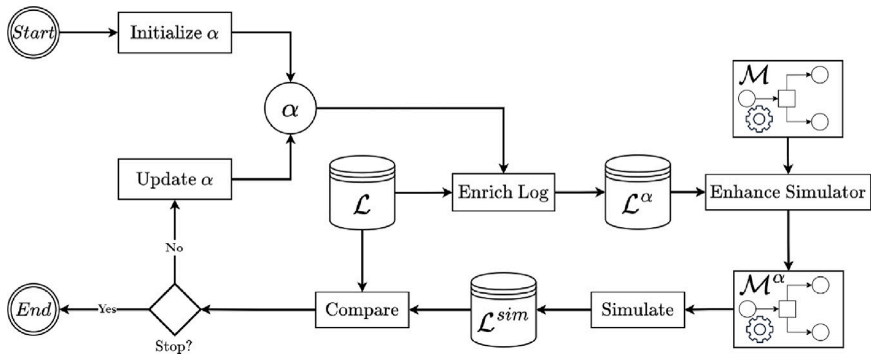

Fig. A.6. Start Timestamps Estimator schema. It starts by initializing an alpha parameter for each activity label. Then they are used for enriching the event log $\mathcal{L}$ obtaining an event log $\mathcal{L}^{\alpha}$ with start timestamps. A simulator model $\mathcal{M}$ is enhanced using the temporal information derived from $\mathcal{L}^{\alpha}$, and then it generates a simulated event log $\mathcal{L}^{sim}$. The simulated event log is compared with the actual one, and finally a stopping criterion determines whether to stop the procedure or update the alpha parameters and proceed in an iterative way.

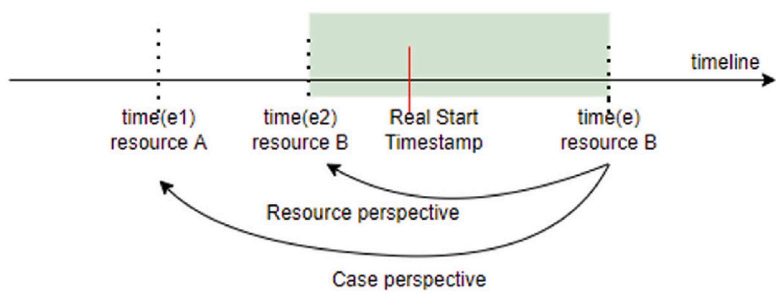

Fig. A.7. Schema representing the concepts of previous event in a trace by control-flow and previous event performed by a resource. For the event e, $time(e)$ is the timestamp related to the event e, and $res(e) = resource\ B$ the resource performing the activity $act(e)$. Looking at the trace perspective, e_1 is the previous event in the trace by control-flow, and e_2 the previous event performed by $resource\ B$. The green box represents the timeline range in which the event e could be started. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Figure from [31].

simulated event log $\mathcal{L}^{sim}$. Finally, $\mathcal{L}^{sim}$ is compared with $\mathcal{L}$, and the set α is adjusted to minimize defined errors. The procedure is repeated for a number of iterations or until a stopping criterion is reached.

Here we define the errors to minimize as the Wasserstein distance [74] between the activity inter-event time distributions (see Definition 4) of the original event log and the simulated one for each activity $a \in \mathcal{A}$:

$$\epsilon_a(\mathcal{L}, \mathcal{L}^{sim}) = \int_0^{+\infty} \left|\Phi_{\mathcal{L}}(a) - \Phi_{\mathcal{L}^{sim}}(a)\right| \tag{A.3}$$

In practice, this measure is approximated using the observed inter-event times from both event logs.

To find the optimal set of α parameters, we employ a local search-based algorithm, as detailed in Algorithm 1. The algorithm iteratively refines the interval $[0, 1]$ for each $\alpha(a)$, progressively halving it and moving towards the extremity that corresponds to the minimum error until a certain stopping criterion, such as a number of iterations or until the errors do not improve.

In Algorithm 1, $l[a]$ and $r[a]$ represent the left and the right bounds of the interval for each α, initialized to 0 and 1, respectively. The function $\texttt{simulate}(\mathcal{M}, \mathcal{L}, \alpha)$ returns the simulated event log as previously described and illustrated in Fig. A.6. For each $a \in \mathcal{A}$, the errors for the intervals are stored in $err[a][l]$ and $err[a][r]$. The function $argmin(err[a][l], err[a][r])$ returns either $l[a]$ or $r[a]$, for which the correspondent error is minimum. The algorithm iterates until `StoppingCriterion`, returning the optimal configuration α.

A.3. Evaluation

In this Section, we assess the quality of the approach, evaluating it on two case studies for which start timestamps are present in the event logs. In particular, we removed the start events in the event-logs, and we used our technique to generate them. The evaluation is based on comparing the actual ones with those obtained using our technique.

We used Prosimos [75] as a simulator engine, integrating it into our implementation. The method has been developed in Python and is available in an online repository.[1]

The evaluation has been conducted on two distinct case studies with available event logs: a synthetic purchasing example process (P2P) and a manufacturing production process (MP). The event logs and their correspondent models are retrieved from the Prosimos GitHub repository.[2]

We compared the starting timestamps obtained by our framework with the actual ones computing the Mean Absolute Errors (MAE) in time units between all the events. To mitigate the stochasticity inherent

Table A.8
Start timestamp estimator results for each case study and method. The column “MAE (min)” indicates the Mean Absolute Error (in min) between the estimated and the actual timestamps. The column “Comp. Time (min)” refers to the computational time. Best results are highlighted in bold.

Case study	Method	MAE (min)	Comp. Time (min)
P2P	Our	**184.24**	**2.03**
	[31]	316.14	8.18
MP	Our	**218.64**	**1.39**
	[31]	374.74	8.54

[1] https://github.com/Franc3sc4/StartTimestampEstimator.
[2] https://github.com/AutomatedProcessImprovement/Prosimos.

Algorithm 1 Start Timestamp Estimator

Require: $\mathcal{L}$: event log; $\mathcal{M}$: simulator model
  for $a \in \mathcal{A}$ **do**
    $l[a] \leftarrow 0$
    $\alpha[a] \leftarrow l[a]$
  end for
  $\mathcal{L}^{sim} \leftarrow \texttt{simulate}(\mathcal{M}, \mathcal{L}, \alpha)$
  for $a \in \mathcal{A}$ **do**
    $err[a][l] \leftarrow \epsilon_a(\mathcal{L}, \mathcal{L}^{sim})$
  end for
  for $a \in \mathcal{A}$ **do**
    $r[a] \leftarrow 1$
    $\alpha[a] \leftarrow r[a]$
  end for
  $\mathcal{L}^{sim} \leftarrow \texttt{simulate}(\mathcal{M}, \mathcal{L}, \alpha)$
  for $a \in \mathcal{A}$ **do**
    $err[a][r] \leftarrow \epsilon_a(\mathcal{L}, \mathcal{L}^{sim})$
  end for
  while StoppingCriterion **do**
    for $a \in \mathcal{A}$ **do**
      $\alpha[a] \leftarrow (l[a] + r[a])/2$
    end for
    $\mathcal{L}^{sim} \leftarrow \texttt{simulate}(\mathcal{M}, \mathcal{L}, \alpha)$
    for $a \in \mathcal{A}$ **do**
      $\overline{\alpha} \leftarrow argmin(err[a][l], err[a][r])$
      if $\overline{\alpha} = l[a]$ **then**
        $r[a] \leftarrow \alpha[a]$
        $err[a][r] \leftarrow \epsilon_a(\mathcal{L}, \mathcal{L}^{sim})$
      else
        $l[a] \leftarrow \alpha[a]$
        $err[a][l] \leftarrow \epsilon_a(\mathcal{L}, \mathcal{L}^{sim})$
      end if
    end for
  end while
  return α

in simulations, we ran the experiments 30 times and averaged the results. Table A.8 reports the outcomes obtained using our approach comparing them with those obtained by Fracca et al. [31].

In addition to MAE, we also evaluated the computational time. Both methodologies were executed on the same machine to ensure a fair comparison, showing that our methodology yields better results and significantly shorter computing time. Specifically, we used an equipment with an AMD Ryzen 7 PRO 3700U w/ Radeon Vega Mobile Gfx 2.30 GHz and 16 GB RAM.

In the P2P case study, our method achieved a 41.72% reduction in MAE and a 75.18% decrease in computational time. Similarly, MP reveals a decrease in MAE of 41.66% and an improvement in computational time of 83.72%. These results suggest the consistency and efficiency of our method. Note that the MAE magnitudes are significant for this kind of process which presents long waiting and execution times.

A.4. Discussion

In this Appendix, we described and showed the efficacy of our proposal technique for repairing missing starting event timestamps. The evaluation indicates promising results, showing that our approach outperforms the previous method proposed by Fracca et al. [31]. A key distinction of our method lies in the optimization function we employ, which is tailored to specific activities (see Eq. (A.3)). Also, unlike the approach by Fracca et al. [31], where activities alpha are changed one at a time, our method changes them in parallel. This permits increasing computational time, and also removing dependencies on previous optimized activities alpha, resulting in faster and more robust solutions. Finally, our method improves the research of alphas by not limiting the search around a starting point depending on a granularity parameter, as done in [31]. In fact, we halve the interval $[0, 1]$ during the search process, leading to better local optima.

The efficacy and accuracy of our approach suggest its potential usability for enriching event logs with missing timestamps. This issue is particularly common in event data collected by Healthcare Information Systems. In this paper, we used this method to address this challenge, showing its application in a real life case study discussed in Section 5.

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